



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians with credentials"
- "Validate password strength"
- "Authenticate librarians and manage access"
- "Validate active sessions"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books in collection"
- "Delete registered books"
- "Consult books by identifier or title"
- "List registered books"
- "List books available for loan"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans"
- "Delete registered readers"
- "Consult readers by identifier or name"
- "List registered readers"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans for readers"

- "Record loan dates"
- "Validate reader and book eligibility for loans"
- "Ensure books are loaned to one reader at a time"
- "Remove loan links for deleted entities"

dependencies:

- "Book Catalog Service"
- "Reader Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/loan-management-service"

- name: "Environment Management Service"

responsibilities:

- "Clear data and base structure to reset environment"
- "Recreate base structure"
- "Coordinate functional areas to restore environment consistency"

dependencies:

- "Authentication Service"
- "Book Catalog Service"
- "Reader Management Service"
- "Loan Management Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Sum integer values"
- "Calculate Fibonacci sequence values"
- "Calculate Ackermann function results"
- "Provide service catalog for testing endpoints"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Record relevant operations from functional flows centrally"
- "Support traceability and problem analysis across domains"

dependencies:

- ""

suggested_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"

to: "Audit Service"

- from: "Book Catalog Service"
to: "Loan Management Service"

- from: "Book Catalog Service"
to: "Audit Service"

- from: "Reader Management Service"
to: "Loan Management Service"

- from: "Reader Management Service"
to: "Audit Service"

- from: "Loan Management Service"
to: "Book Catalog Service"

- from: "Loan Management Service"
to: "Reader Management Service"

- from: "Loan Management Service"
to: "Audit Service"

- from: "Environment Management Service"
to: "Authentication Service"

- from: "Environment Management Service"
to: "Book Catalog Service"

- from: "Environment Management Service"
to: "Reader Management Service"

- from: "Environment Management Service"
to: "Loan Management Service"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customer login and password"
- "Create and manage customer sessions"
- "Validate active sessions"
- "Process customer logout"

dependencies:

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suggested_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Maintain registered airports and routes"
- "Search one-way and round-trip flights"
- "Provide flight schedules and reference fares"

dependencies:

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suggested_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create one-way and round-trip bookings"
- "Consult and list customer bookings"
- "Process booking cancellations"

dependencies:

- "Authentication Service"
- "Flight Search Service"

suggested_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Retrieve customer profile data"
- "Update customer contact and address data"
- "Verify password for profile updates"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Present customer loyalty status"
- "Display annual and total accumulated miles"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Load and reload application database data"
- "Generate sample operational records"
- "Provide summarized operational indicators and metrics"

dependencies:

- "Authentication Service"
- "Flight Search Service"
- "Booking Service"
- "Customer Profile Service"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"

to: "Authentication Service"

- from: "Booking Service"

to: "Flight Search Service"

- from: "Customer Profile Service"

to: "Authentication Service"

- from: "Loyalty Service"

to: "Authentication Service"

- from: "Environment Administration Service"

to: "Authentication Service"

- from: "Environment Administration Service"

to: "Flight Search Service"

- from: "Environment Administration Service"

to: "Booking Service"

- from: "Environment Administration Service"

to: "Customer Profile Service"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Architecture A and Architecture B have identical Service metrics (Precision=1.0000"

responsibilities:

- "Recall=0.8333"

dependencies:

- "F1=0.9091)."

suggested_endpoints:

- "/api/architecture-a-and-architecture-b-have-identical-service-metrics-(precision=1.0000"

- name: "Architecture B has significantly higher Interaction metrics (Precision=1.0000"

responsibilities:

- "Recall=0.7273"

dependencies:

- "F1=0.8421) compared to Architecture A (Precision=0.3333"

suggested_endpoints:

- "/api/architecture-b-has-significantly-higher-interaction-metrics-(precision=1.0000"

- name: "Output must be raw CSV with no markdown or extra text.Microservice"

responsibilities:

- "Responsibilities"

dependencies:

- "Communicates With"

suggested_endpoints:

- "/api/output-must-be-raw-csv-with-no-markdown-or-extra-text.microservice"

- name: "Cargo Management Service"

responsibilities:

- "Registering new cargo with origin destination and arrival deadline"

- "Assigning unique tracking identifier to cargo"
- "Managing cargo lifecycle and routing status"
- "Updating cargo destination and triggering route reevaluation"

dependencies:

- "Route Planning Service"

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Retrieving and presenting candidate routes compatible with cargo needs"
- "Validating candidate routes against origin destination and deadline"
- "Displaying route legs load points unload points voyages and expected dates"
- "Replanning routes for off-route or modified cargo"

dependencies:

- "Cargo Management Service"

suggested_endpoints:

- "/api/route-planning-service"

- name: "Logistics Event Registration Service"

responsibilities:

- "Registering logistics events including reception load unload customs and pickup"
- "Validating event occurrence location cargo and voyage"
- "Rejecting invalid event registration attempts"
- "Updating cargo operational situation and recalculating transport status and plan adherence"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"

suggested_endpoints:

- "/api/logistics-event-registration-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Consulting cargo current situation by tracking identifier"
- "Consolidating cargo operational status and logistics event history"
- "Tracking last known location and calculating estimated arrival"
- "Identifying off-route status and providing geographic tracking visualization"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Logistics Event Registration Service"

suggested_endpoints:

- "/api/cargo-tracking-service"

- name: "Integration Service"

responsibilities:

- "Receiving logistics events across operational interface integration service and batch file processing"

- "Processing input files periodically and isolating invalid records"
- "Providing consolidated cargo view for monitoring channels"
- "Exchanging standardized cargo tracking information with external systems"

dependencies:

- "Logistics Event Registration Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/integration-service"

interactions:

- from: "Architecture A and Architecture B have identical Service metrics (Precision=1.0000"
to: "F1=0.9091)."
- from: "Architecture B has significantly higher Interaction metrics (Precision=1.0000"
to: "F1=0.8421) compared to Architecture A (Precision=0.3333"
- from: "Output must be raw CSV with no markdown or extra text.Microservice"
to: "Communicates With"
- from: "Cargo Management Service"
to: "Route Planning Service"
- from: "Route Planning Service"
to: "Cargo Management Service"
- from: "Logistics Event Registration Service"
to: "Cargo Management Service"
- from: "Logistics Event Registration Service"
to: "Route Planning Service"
- from: "Cargo Tracking Service"
to: "Cargo Management Service"
- from: "Cargo Tracking Service"
to: "Route Planning Service"
- from: "Cargo Tracking Service"
to: "Logistics Event Registration Service"
- from: "Integration Service"
to: "Logistics Event Registration Service"
- from: "Integration Service"
to: "Cargo Tracking Service"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate investors with username and password"
- "Manage active investor sessions"
- "Terminate investor sessions through logout"
- "Prevent unauthorized access to protected resources"

dependencies:

- "Investor Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Investor Account Service"

responsibilities:

- "Register new investors"
- "Validate investor registration and password confirmation"
- "Maintain investor accounts and financial data"
- "Record initial balance and current balance"
- "Record access dates"

dependencies:

- "total logins"

suggested_endpoints:

- "/api/investor-account-service"

- name: "Market Data Service"

responsibilities:

- "Maintain asset quote information"
- "Provide individual and simultaneous quote queries"
- "Handle invalid asset codes without navigation interruption"
- "Generate consolidated market performance summary"
- "Update traded volumes and market indicators following trading operations"

dependencies:

- ""

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Manage lifecycle of buy and sell orders"
- "Generate trading orders with status tracking and fee calculation"
- "Validate asset position availability before processing sales"
- "Coordinate balance debit and credit with investor accounts"
- "Coordinate position addition and removal with investor portfolio"
- "Propagate volume and indicator updates to market data"
- "Maintain investor order history"
- "Alert investors about completed orders"
- "Obtain simulation processing configurations"

dependencies:

- "Investor Account Service"
- "Market Data Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/trading-service"
- name: "Portfolio Service"
- responsibilities:
 - "Maintain investor asset positions"
 - "Add new positions from completed buy orders"
 - "Remove positions resulting from completed sell orders"
 - "Calculate position market value and gain or loss using current market quotes"
 - "Calculate consolidated financial position summary using cash balance"
 - "Allow selling positions directly from the portfolio"

dependencies:

- "Investor Account Service"
- "Market Data Service"
- "Trading Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters"
- "Configure order processing mode"

dependencies:

- "simulation limits"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"
to: "Investor Account Service"
- from: "Investor Account Service"
to: "total logins"
- from: "Trading Service"
to: "Investor Account Service"
- from: "Trading Service"
to: "Market Data Service"
- from: "Trading Service"
to: "Portfolio Service"
- from: "Trading Service"
to: "Simulation Administration Service"
- from: "Portfolio Service"
to: "Investor Account Service"
- from: "Portfolio Service"
to: "Market Data Service"
- from: "Portfolio Service"
to: "Trading Service"
- from: "Simulation Administration Service"

to: "simulation limits"

SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users with username and password"
- "Authenticate registered users"
- "Manage user sessions"
- "Control and differentiate user profiles"
- "Prevent duplicate user registration"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "List available movies with pagination"
- "Provide movie details and full cast"
- "Manage movie categories"
- "Filter movies by category"
- "Register new movies in catalog"
- "Delete movies individually and in batch"
- "Prevent duplicate movie registration"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Upload movie video files"
- "Store video files"
- "Deliver video content for playback"
- "Provide movie download links"
- "Verify video file availability in delivery area and local repository"

dependencies:

- "Authentication Service"
- "Catalog Service"

suggested_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"

to: "Authentication Service"

- from: "Media Service"

to: "Authentication Service"

- from: "Media Service"

to: "Catalog Service"

SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Organize products by animal categories"
- "Manage product and item details"
- "Provide keyword search for products"
- "Display item pricing and stock status"

dependencies:

- "Inventory Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Customer Account Service"

responsibilities:

- "Register new customers"
- "Manage customer profile and registration data"
- "Manage user preferences and favorite categories"
- "Enable personalized category views and promotional banners"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-account-service"

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Manage user login sessions"
- "Handle user logout and session termination"
- "Provide access control for protected features"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Shopping Cart Service"

responsibilities:

- "Manage temporary item selection"
- "Update item quantities and remove items"

- "Calculate cart subtotals and totals"
- "Verify item stock availability at inclusion"

dependencies:

- "Catalog Service"
- "Inventory Service"

suggested_endpoints:

- "/api/shopping-cart-service"

- name: "Inventory Service"

responsibilities:

- "Maintain item stock levels"
- "Provide stock availability and backorder status"
- "Register stock reductions upon order completion"

dependencies:

- "Catalog Service"

suggested_endpoints:

- "/api/inventory-service"

- name: "Order Service"

responsibilities:

- "Coordinate checkout process"
- "Retrieve customer billing and delivery addresses"
- "Assemble order details from shopping cart"
- "Capture payment details"
- "Finalize orders with unique purchase identifiers"
- "Record order history and lifecycle status"
- "Clear shopping cart after purchase completion"
- "Allow consultation of completed orders"

dependencies:

- "Authentication Service"
- "Customer Account Service"
- "Inventory Service"
- "Shopping Cart Service"

suggested_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"
to: "Inventory Service"
- from: "Customer Account Service"
to: "Authentication Service"
- from: "Shopping Cart Service"
to: "Catalog Service"
- from: "Shopping Cart Service"
to: "Inventory Service"
- from: "Inventory Service"

to: "Catalog Service"

- from: "Order Service"

to: "Authentication Service"

- from: "Order Service"

to: "Customer Account Service"

- from: "Order Service"

to: "Inventory Service"

- from: "Order Service"

to: "Shopping Cart Service"

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owner profiles"
- "Update registered owner data"
- "Search owners by last name or broad criteria"
- "Retrieve detailed owner information"

dependencies:

- ""

suggested_endpoints:

- "/api/owner-service"

- name: "Animal Service"

responsibilities:

- "Register animals linked to owners"
- "Update registered animal data"
- "Provide predefined list of animal types"
- "Validate animal birth date and name uniqueness per owner"

dependencies:

- "Owner Service"

suggested_endpoints:

- "/api/animal-service"

- name: "Visit Service"

responsibilities:

- "Register visits for specific animals"
- "Maintain animal visit and service history"
- "Associate visits with animal and owner history"
- "Resume pending visit processes after restart"

dependencies:

- "Animal Service"
- "Veterinarian Service"

suggested_endpoints:

- "/api/visit-service"

- name: "Veterinarian Service"

responsibilities:

- "List veterinarians with specialties and next expected visits"
- "Automatically assign visits based on lowest future workload"
- "Calculate veterinarian workload and next visits"
- "Provide structured veterinarian data for external consumption"

dependencies:

- "Visit Service"

suggested_endpoints:

- "/api/veterinarian-service"

interactions:

- from: "Animal Service"
to: "Owner Service"
- from: "Visit Service"
to: "Animal Service"
- from: "Visit Service"
to: "Veterinarian Service"
- from: "Veterinarian Service"
to: "Visit Service"

SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration and Configuration Service"

responsibilities:

- "Maintain support records"
- "Manage user configurations"
- "Provide system configuration parameters"

dependencies:

- ""

suggested_endpoints:

- "/api/administration-and-configuration-service"

- name: "Identity and Access Service"

responsibilities:

- "Register and maintain user data"
- "Manage passwords and authentication"
- "Control access by profile and scope"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/identity-and-access-service"

- name: "Workforce Management Service"

responsibilities:

- "Maintain work agreements"
- "Manage vacation requests and balances"
- "Send workforce notifications"

dependencies:

- "Administration and Configuration Service"
- "Identity and Access Service"
- "Notification Service"

suggested_endpoints:

- "/api/workforce-management-service"

- name: "Project Tracking Service"

responsibilities:

- "Maintain client projects"
- "Register internal and external activities"
- "Calculate worked hours"
- "Enforce evidence policies"

dependencies:

- "File Storage Service"
- "Identity and Access Service"
- "Notification Service"
- "Workforce Management Service"

suggested_endpoints:

- "/api/project-tracking-service"

- name: "CRM and Proposals Service"

responsibilities:

- "Maintain organizations and contacts"
- "Manage client interactions"
- "Manage proposals and proposal states"

dependencies:

- "Administration and Configuration Service"
- "Project Tracking Service"

suggested_endpoints:

- "/api/crm-and-proposals-service"

- name: "Financial Management Service"

responsibilities:

- "Maintain financial accounts and entries"
- "Calculate financial indicators"
- "Import billable hours and costs"

dependencies:

- "Billing Service"
- "Project Tracking Service"

suggested_endpoints:

- "/api/financial-management-service"

- name: "Billing Service"

responsibilities:

- "Detail invoices and calculate totals"
- "Calculate taxes and balances"
- "Process proposal conversions and payments"

dependencies:

- "Administration and Configuration Service"
- "CRM and Proposals Service"
- "File Storage Service"
- "Financial Management Service"
- "Project Tracking Service"

suggested_endpoints:

- "/api/billing-service"

- name: "Content and Document Management Service"

responsibilities:

- "Organize documents by category"
- "Manage document versions"
- "Associate documents with users and areas"

dependencies:

- "File Storage Service"
- "Identity and Access Service"

suggested_endpoints:

- "/api/content-and-document-management-service"

- name: "Commissioning Service"

responsibilities:

- "Register and manage orders and commissions"
- "Track commission execution"

dependencies:

- "File Storage Service"
- "Notification Service"
- "Project Tracking Service"
- "Workforce Management Service"

suggested_endpoints:

- "/api/commissioning-service"

- name: "Reporting Service"

responsibilities:

- "Consolidate business reports"
- "Export domain reports"

dependencies:

- "Billing Service"
- "Commissioning Service"
- "CRM and Proposals Service"
- "Financial Management Service"

- "Project Tracking Service"
- "Workforce Management Service"

suggested_endpoints:

- "/api/reporting-service"

- name: "Notification Service"

responsibilities:

- "Send email notifications"
- "Manage notification settings"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/notification-service"

- name: "File Storage Service"

responsibilities:

- "Store and retrieve files"
- "Manage storage configurations"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/file-storage-service"

interactions:

- from: "Identity and Access Service"
to: "Administration and Configuration Service"
- from: "Workforce Management Service"
to: "Administration and Configuration Service"
- from: "Workforce Management Service"
to: "Identity and Access Service"
- from: "Workforce Management Service"
to: "Notification Service"
- from: "Project Tracking Service"
to: "File Storage Service"
- from: "Project Tracking Service"
to: "Identity and Access Service"
- from: "Project Tracking Service"
to: "Notification Service"
- from: "Project Tracking Service"
to: "Workforce Management Service"
- from: "CRM and Proposals Service"
to: "Administration and Configuration Service"
- from: "CRM and Proposals Service"
to: "Project Tracking Service"
- from: "Financial Management Service"
to: "Billing Service"

- from: "Financial Management Service"
to: "Project Tracking Service"

- from: "Billing Service"
to: "Administration and Configuration Service"

- from: "Billing Service"
to: "CRM and Proposals Service"

- from: "Billing Service"
to: "File Storage Service"

- from: "Billing Service"
to: "Financial Management Service"

- from: "Billing Service"
to: "Project Tracking Service"

- from: "Content and Document Management Service"
to: "File Storage Service"

- from: "Content and Document Management Service"
to: "Identity and Access Service"

- from: "Commissioning Service"
to: "File Storage Service"

- from: "Commissioning Service"
to: "Notification Service"

- from: "Commissioning Service"
to: "Project Tracking Service"

- from: "Commissioning Service"
to: "Workforce Management Service"

- from: "Reporting Service"
to: "Billing Service"

- from: "Reporting Service"
to: "Commissioning Service"

- from: "Reporting Service"
to: "CRM and Proposals Service"

- from: "Reporting Service"
to: "Financial Management Service"

- from: "Reporting Service"
to: "Project Tracking Service"

- from: "Reporting Service"
to: "Workforce Management Service"

- from: "Notification Service"
to: "Administration and Configuration Service"

- from: "File Storage Service"
to: "Administration and Configuration Service"